

Devendra Garwa

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 GitHub |  LinkedIn |  Portfolio

EDUCATION

- Indian Institute of Technology Roorkee 2021 – 2026
Integrated M.Tech, Geological Technology
- Matrix High School, Sikar 2020
Class XII | RBSE
- Euro International School, Sikar 2018
Class X | CBSE

SKILLS

Programming Languages: Python, SQL

Libraries & Packages: Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn

Visualization & BI Tools: Power BI, Tableau, Plotly

Databases & Tools: MySQL, Git, GitHub, Jupyter Notebook, Google Colab, MS Excel

Techniques: Exploratory Data Analysis, Feature Engineering, Classification, Regression, Risk Scoring

EXPERIENCE

Machine Learning Engineer Intern | Technocolabs Softwares Inc. Jun 2025 – Jul 2025

- Analyzed **35K+** delivery records to detect logistics inefficiencies and improve delivery-time predictability.
- Engineered **20+** domain-specific features on route efficiency and delay patterns for delivery **ETA** prediction.
- Built a **CatBoost** regression model to predict delivery **ETA**, validated with **MAE** and **R²** on a held-out test set.

PROJECTS

Credit Card Default Prediction Using Classification and Risk-Based Techniques May 2025 – Jun 2025

- Conducted **EDA** on **25K+** credit card records with **30+** financial, behavioral, and demographic features for risk analysis.
- Trained an **XGBoost** classifier with **hyperparameter tuning**, achieving **0.72 F2-score** and **75%** recall.
- Employed **SMOTE** for class imbalance and calibrated a **0.27 decision threshold** for high-risk default detection.

A Data-Driven Approach to Flight Delay Analysis and Prediction Jun 2024

- Performed **EDA** on **30K+** US flight records across **150+** airports, uncovering seasonal delay patterns and root causes.
- Developed a **Random Forest** model for delay **classification (0.94 AUC)** and duration **regression (6.3 min MAE)**.
- Applied **SHAP** to identify carrier and late-aircraft delays as key controllable factors guiding operational improvements.

Oil & Gas Zone Detection Using Seismic Attenuation Analysis 2025 – 2026

- Constructed an end-to-end **Python** pipeline to process **~170 km** of seismic survey data for attenuation analysis.
- Implemented **Spectral Ratio** and **Centroid Frequency Shift** methods, cross-validated to **<2%** cross-validation error.
- Isolated validated **anomaly clusters** flagging candidate hydrocarbon zones, visualized through interactive **3D** maps.

Solving Inventory Inefficiencies Using Advanced SQL Analytics Aug 2025

- Designed a structured **MySQL** database for **5,000+** SKUs to centralize inventory data and improve analytical accessibility.
- Executed **SQL-based analytics** and **anomaly detection** on key inventory **KPIs** to flag overstock and stockout risks.
- Built **Power BI** dashboards to visualize SKU, category, and regional inventory trends for actionable insights.

Customer Churn Prediction and Retention Strategy for Subscription Services May 2024 – Jun 2024

- Trained a **Random Forest** classifier on **7,000+** telecom customers to predict churn via supervised **classification**.
- Conducted **EDA** and built a preprocessing pipeline for encoding, missing-value treatment, and **feature engineering**.
- Utilized **SMOTE-ENN** resampling with **hyperparameter tuning** to significantly improve minority-class recall.

LEADERSHIP & EXTRACURRICULARS

Joint Secretary, Public Relations | Rasaionix, IIT Roorkee Jul 2024 – Jun 2025

- Led PR campaigns across campus and social channels to increase event attendance and stakeholder engagement.
- Oversaw content calendar, media partnerships, and analytics to optimize messaging and outreach performance.

Executive Member | Himalayan Explorers' Club, IIT Roorkee Jun 2023 – May 2024

- Organized logistics and outreach for **10+** trekking expeditions, fostering strong team participation and engagement.
- Delivered safety training to **100+** students, enhancing expedition readiness and boosting trek participation by **30%**.

CERTIFICATIONS & COURSES

Google Data Analytics (Coursera) | Data Analytics Job Simulation — Forage (Quantium) | WorldQuant BRAIN Challenge — Bronze (2022) | Convolve 4.0 — pan-IIT AI/ML Hackathon (2026) | Winter Consulting — C&A Club, IIT Guwahati | Google Project Management (Coursera)